

The Wind of Change: Maritime Technology, Trade, and Economic Development[†]

By LUIGI PASCALI*

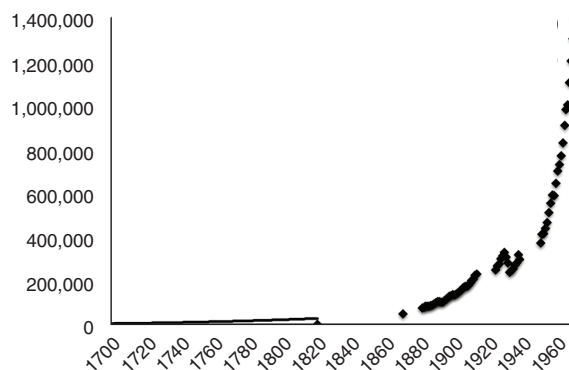
The 1870–1913 period marked the birth of the first era of trade globalization. How did this tremendous increase in trade affect economic development? This work isolates a causality channel by exploiting the fact that the introduction of the steamship in the shipping industry produced an asymmetric change in trade distances among countries. Before this invention, trade routes depended on wind patterns. The steamship reduced shipping costs and time in a disproportionate manner across countries and trade routes. Using this source of variation and novel data on shipping, trade, and development, I find that (i) the adoption of the steamship had a major impact on patterns of trade worldwide; (ii) only a small number of countries, characterized by more inclusive institutions, benefited from trade integration; and (iii) globalization was the major driver of the economic divergence between the rich and the poor portions of the world in the years 1850–1900. (JEL F14, F43, F63, L92, N70, O33)

The 1870–1913 period marked the birth of the first era of trade globalization. As shown in Figure 1, between 1820 and 1913, the world experienced an unprecedented increase in world trade, with a marked acceleration that began in 1870. This increase in trade cannot simply be explained by increased global GDP or population. In fact, between 1870 and 1913, the world export-to-GDP ratio increased from 5 percent to 9 percent, while per capita volumes more than tripled. The determinants and consequences of this first wave of globalization have been of substantial interest to both economists and historians. This study employs new trade data and a novel identification strategy to empirically investigate (i) the role of the adoption of the

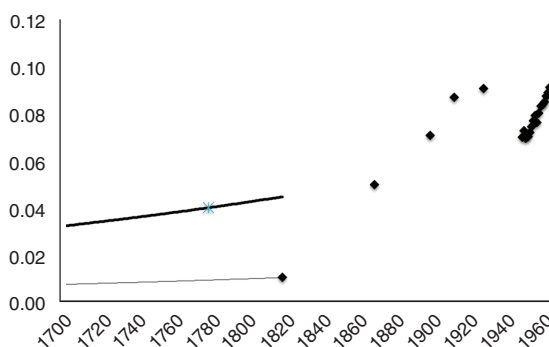
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Panel A. World exports at constant prices (million 1990 dollars)



Panel B. World export-to-GDP ratio



Panel C. World export-to-population ratio

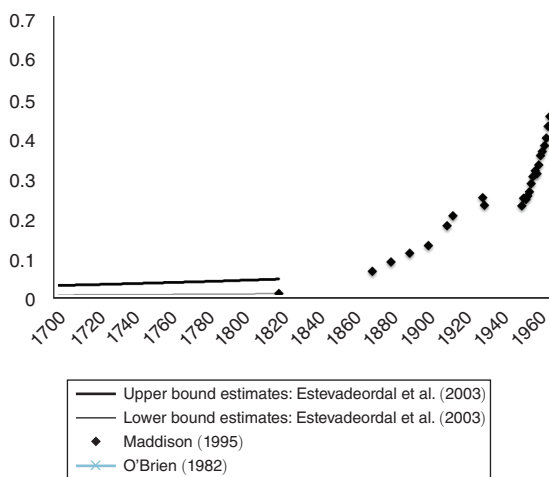


FIGURE 1. WORLD TRADE FROM 1700 TO 1970

Notes: The “lower bound” and “upper bound” series from 1700 to 1820 on export share (reported in panel B) come from Estevadeordal, Frantz, and Taylor (2003). In panels A and C, these series were rescaled by the author using data on world GDP and population (due to Maddison 2004) to obtain lower and upper bound series on total export and export-to-population ratio from 1700 to 1820.

steamship in shaping the pattern of world trade in the nineteenth century and (ii) the effect of this tremendous increase in world trade on economic development.

This paper isolates a causality channel by exploiting the fact that the steamship produced an asymmetric change in shipping times across countries. Before the invention of the steamship, trade routes depended on wind patterns. The adoption of the steam engine reduced shipping times in a disproportionate manner across countries and trade routes. For instance, because the winds in the Northern Atlantic Ocean follow a clockwise pattern, the duration of a round trip for a clipper ship from Lisbon to Cape Verde would be similar to that of a round trip from Lisbon to Salvador. However, with the steamship, the former trip would require only half of the time needed for the latter trip. These asymmetric changes in shipping times across countries are used to identify the effect of the adoption of the steamship on trade patterns and volumes and to explore the effect of international trade on economic development.

This paper is based on a large collection of data, as it uses three completely novel datasets that span the great majority of the world from 1850 to 1900. The first dataset provides information on shipping times using different sailing technologies across approximately 16,000 country pairs. The second dataset consists of more than 23,000 bilateral trade observations for nearly 1,000 distinct country pairs and sectoral-level export data pertaining to 37 countries. Finally, the third dataset provides information on freight rates across 291 major shipping routes. These data are then combined with a large dataset on logbooks of sailing voyages between the eighteenth and the nineteenth centuries and more traditional resources on per capita income, population, and urbanization rates.

Four key findings emerge from this analysis.

First, regressions of bilateral trade on shipping times by both sail and steam vessels between 1850 and 1900 reveal that trade patterns were shaped by shipping times by sail until 1865, by a weighted average of shipping times by sail and steam between 1866 and 1875, and by shipping times by steam thereafter. I also document that this shift in patterns of trade was the result of a change in relative freight rates induced by the steamship.

Second, I provide a rough estimate of the impact of the steamship on world trade volumes. I measure the geographical isolation of a country as the average shipping time from this country to the remainder of the world, and I use country-level regressions to estimate the impact of the change in isolation, induced by the steamship, on the change in trade volumes between 1850 and 1905. The estimated elasticity is surprisingly large. Using a simple back-of-the-envelope calculation, I argue that the reduction in shipping times induced by the steam engine might be responsible for approximately half of the increase in international trade during the second half of the nineteenth century. Admittedly this is a very crude estimate and it should be taken with a grain of salt, as it is based on the assumption that the rollout of the steam was uniform across different countries and different products and it was completely concluded by the end of the period of analysis.

Third, the predictions for bilateral trade generated by the regressions of trade on shipping times by sail and steam vessels are then summed to generate a panel of overall trade predictions for 37 countries from 1850 to 1905 (I limit the analysis to countries for which data on total exports and per capita GDP are available both before and after the year 1870). These predictions can be used as instruments in

panel regressions of trade on per capita income, population, and urbanization rates, and the time series variation of the instrument allows for the inclusion of time- and country-specific fixed effects in the second-stage regression. I find that the effect of trade on development and urbanization is not necessarily positive: the average impact, in the short run, of the first wave of trade globalization was a reduction in per capita GDP, population, and urbanization rates all over the world. This average effect, however, masks large differences across groups of countries. In particular, an exogenous increase in international trade produced different effects depending on the initial levels of economic development: it was highly detrimental in countries characterized by a per capita GDP below the mean in 1850, while it had a very limited impact on the economic performance of the richest countries. Using these estimates, I argue that the majority of the economic divergence that is observed between the richest countries and the rest of the world in the second half of the nineteenth century can be attributed to the first wave of trade globalization.

Finally, I find that the effect of trade on economic development is beneficial for countries that are characterized by strong constraints on executive power, which is a distinct feature of the institutional environment that has been demonstrated to favor private investment. Following the first wave of trade globalization, these countries specialized in nonagricultural products and benefited from trade. More specifically, an exogenous doubling in the export-to-GDP ratio reduced per capita GDP growth rates by more than a third in countries characterized by an executive power with unlimited authority, while it increased per capita GDP growth rates by almost one-tenth in countries in which the executive power was obliged to respond to several accountability groups. Moreover, the latter countries increased the share of their exports in nonmanufacturing goods by almost 12 percent (while this share did not significantly change for the rest of the sample). This result is relevant to the large stream of literature that has argued that institutions are crucial to obtaining benefits from international trade, although I still view it as preliminary and exploratory: with a sample of only 37 countries, it is indeed very difficult to pin down the precise channels and mechanisms through which trade affects development with a reasonable degree of certainty.

To the best of my knowledge, this study is the first work that quantifies the effects of the adoption of the steamship on global trade volumes and economic development in a well-identified empirical framework. This work also contributes to several strands of the economic literature.

First, my findings contribute to the debate on the importance of reduced transportation costs in spurring international trade during the first wave of globalization. The most widely held perspective on the nineteenth century is that while railroads were responsible for promoting within-country integration, steamships served the same role in promoting cross-country integration (Frieden 2007; James 2001). However, this view is not reflected in the most recent empirical literature examining the first wave of trade globalization (O'Rourke and Williamson 1999; Estevadeordal, Frantz, and Taylor 2003; Jacks, Meissner, and Novy 2011). In particular, these studies have emphasized the role of income growth when focusing on explaining the increase in absolute trade and the role of the combination of decreasing transportation costs and the adoption of the gold standard when focusing on trade shares. The typical methodology applied in this stream of literature is to regress exports on freight

rates and then to calculate the share of the increase in trade after 1870 that can be explained by the contemporaneous reduction in freight rates. Thus, my paper addresses a major identification issue: freight rates are endogenous because they are likely to be affected not only by technology but also by changes in economic activity or market structure. Additionally, my work is the first to extend the period of analysis before 1870. This extended period is necessary to capture the transition period from sail to steam vessels and to explain the sources of the structural break in trade data after 1870.

Second, my findings contribute to the debate on the effects of trade on development. Although I am not aware of any paper that identifies a causal link in the nineteenth century, a large body of literature has focused on more recent years. Beginning with the seminal work of Frankel and Romer (1999), a large number of papers have attempted to identify a causal channel using a geographic instrument: the point-to-point great circle distance across countries. Although this instrument is free of reverse causality, it is correlated with geographic differences in outcomes that are not generated through trade. For instance, countries that are closer to the equator generally have longer trade routes and may have low incomes because of unfavorable disease environments or unproductive colonial institutions. Rodriguez and Rodrik (2001) and others have demonstrated that Frankel and Romer's results are not robust to the inclusion of geographic controls in the second stage. More recently, Feyrer (2009a, 2009b) exploit two natural experiments: the closing of the Suez Canal between 1967 and 1975 and improvements in aircraft technology that generated asymmetric shocks in trade distances. Feyrer finds that an increase in trade exerts large positive effects on economic development. My work demonstrates that although trade has been proven to exert generally positive effects on development in the present day, this might have not been the case one century ago.

Third, my findings contribute to the theoretical debate between neoclassical trade theories, in which comparative advantages are determined by technological differences and factor endowments, and new economic geography theories, in which countries derive part of their comparative advantage from scale economies. Trade liberalization in the conventional Ricardian or Heckscher-Ohlin approach allows countries to exploit their comparative advantage: greater integration may harm particular interest groups but typically increases income in all countries. This view has been challenged by the new economic geography theories (see, for instance, Krugman 1991; Matsuyama 1992; Krugman and Venables 1995; Baldwin, Martin, and Ottaviano 2001; and Crafts and Venables 2003). Although production has constant returns to scale in the neoclassical world, these theories are based on increasing returns within firms and in the economy more broadly. Specifically, production in agriculture is still modeled with constant returns, whereas production in manufacturing shows increasing returns to scale. When trade costs are sufficiently high, a reduction in trade costs together with localized externalities¹ causes a process of industrial agglomeration that is beneficial for countries that specialize in manufacturing and might be detrimental to countries

¹ In Krugman (1991), external economies arise from the desire of firms to establish their facilities close to customers/workers; in Krugman and Venables (1995), externalities arise from linkages between firms; and in Baldwin, Martin, and Ottaviano (2001), externalities arise from capital accumulation in the manufacturing sector.

that specialize in agriculture. My empirical findings support this second stream of literature, as the first wave of trade globalization, in the short run, had positive effects for a small core of countries while exerting negative effects for other countries. A similar empirical result, but limited to Chinese provinces, can be found in Faber (2014) who shows that connections to China's National Truck Highway System (a major investment in highway connections between the major Chinese cities carried out in the period from 1992–2007) led to a reduction in GDP growth among peripheral counties.

Finally, my findings speak to the significant body of empirical literature, beginning with the seminal contributions of Acemoglu, Johnson, and Robinson (2001); Engerman and Sokoloff (1994); and La Porta et al. (1997),² which has convincingly shown that strong institutions (e.g., with respect to shareholder protection, the strength of contract enforcement, and property rights) are critical for economic growth. The closest paper in this sense is Acemoglu, Johnson, and Robinson (2005), which shows that the rise of Atlantic trade between the sixteenth and the nineteenth centuries produced a large positive impact on per capita GDP and urbanization only in those European countries that were characterized by political institutions that placed significant checks on monarchy. Levchenko (2007) demonstrates that, in a model containing differences in contracting imperfections across countries, trade is beneficial in countries with the strongest institutions, while it might become detrimental in countries characterized by weak institutions, which will specialize in those goods that are not “institutionally dependent.” To the best of my knowledge, my work presents the first assessment of this theory, and, together with the work of Acemoglu, Johnson, and Robinson, it provides an empirical basis for an additional channel through which institutions affect economic development.

This paper is structured as follows. Section I provides a description of the evolution of shipping technology during the second half of the nineteenth century. Section II describes the construction of shipping times, trade figures, and the other data used in the paper. Section III describes the effects of the introduction of the steamship in the shipping industry on global patterns and volumes of international trade. Section IV examines the effect of trade on development and urbanization as well as the role of institutions. Some concluding remarks close the paper.

I. From Sail to Steam

The nineteenth century marked an era of spectacular advancements in terms of economic integration throughout the world. It is generally believed that while the construction of new railroads fostered within-country economic integration, the introduction of steam vessels in the shipping industry encouraged cross-country integration. In fact, the great majority of international trade in this period was conducted by sea (see Table A.1 in the online Appendix). The reductions in trade costs between countries, however, were not uniform across trade routes. To illustrate the asymmetric effects on international patterns of trade induced by the shift from sail

² Studies on the effects of history on long-lasting institutions have built upon an earlier body of literature dating back to North and Thomas (1973) and North (1981, 1990). For a complete review, see Nunn (2009).

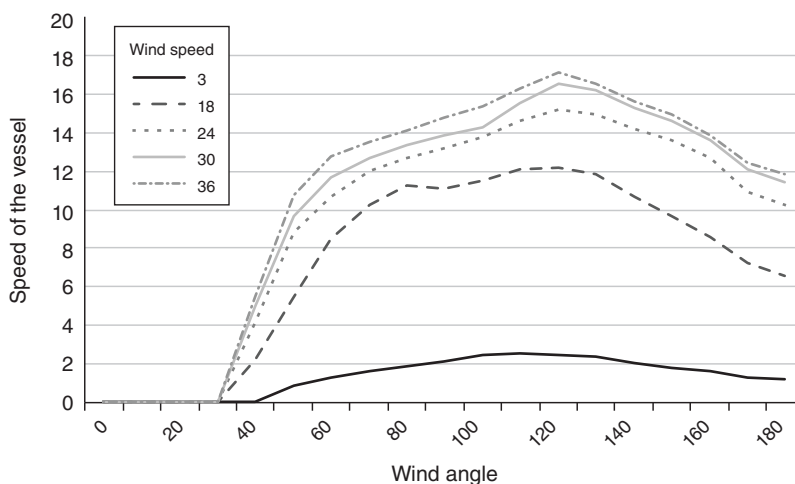


FIGURE 2. POLAR DIAGRAM OF A SAILING VESSEL: THE CLIPPER IN 1860

Note: The polar diagram defines the maximum boat speed achievable for a given wind speed and wind angle.

to steam, in what follows I describe the two competing technologies and their evolution in the second half of the century.

A. The Sailing Vessels

Figure 2 provides a polar diagram of a clipper, a fast-sailing ship that had three or more masts and a square rig and that was largely used for international trade during the nineteenth century. A polar diagram is a compact means of graphing the relationship between the speed of a sailing vessel and the angle and strength of the wind. A clipper cannot navigate against the wind similar to all other sailing vessels, and it reaches its maximum speed when sailing downwind at 140 degrees off the wind. Additionally, the wind speed affects the speed of the vessel, which is maximized when the wind is moving at 24 knots. Given this technology, the prevailing direction and speed of the winds become important determinants shaping the main international trade routes. Figures A.1 and A.2 in the online Appendix present the prevailing wind patterns worldwide and in Europe, while Figure 4 depicts a series of journeys made by British ships between 1800 and 1860 between England, Cape of Good Hope, and Java. For instance, the winds tend to follow a clockwise pattern in the North Atlantic; thus, it is much easier to sail westward from Western Europe after traveling south to 30° N latitude and reaching the “trade winds,” thus arriving in the Caribbean, rather than traveling straight to North America. The result is that trade systems historically tended to follow a triangular pattern among Europe, Africa, the West Indies, and the United States. Furthermore, because the South Atlantic winds tend to blow counterclockwise, British ships would not sail directly southward to the Cape of Good Hope; rather, they would first sail southwest toward Brazil and then move east to the Cape of Good Hope at 30° S latitude.

In summary, given this technology, geographic distances might not be a strong predictor of the trade distance between different ports and countries.

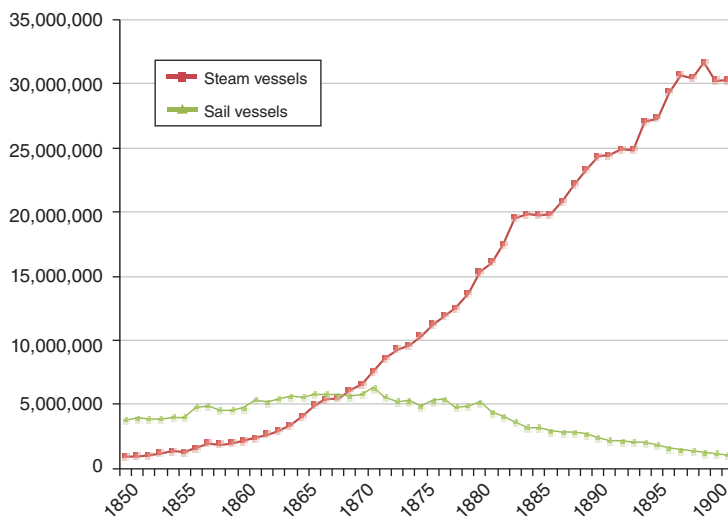


FIGURE 3. TOTAL TONNAGE OF BRITISH VESSELS ENTERED IN BRITISH PORTS FROM AND TO FOREIGN COUNTRIES AND BRITISH POSSESSIONS

Source: *Statistical Abstract* for the United Kingdom (various years from 1851 to 1901)

B. The Steam Vessels

The invention and subsequent development of the steamship represents a watershed event in maritime transport. For the first time, vessels were not at the mercy of the winds, and trade routes became independent of wind patterns.

The first steamship prototypes emerged in the early 1800s. In 1786, John Fitch built the first steamboat, which subsequently operated in regular commercial service along the Delaware River. The early steamboats were small wooden vessels that used low-pressure steam engines and paddle wheels. The paddles were replaced by screw propellers and wooden hulls by iron hulls beginning in the 1840s.

Steam first displaced sail in passenger and intranational trade. Inefficient engines prevented these early steamships from being used in international trade, as longer voyages meant that a greater proportion of a ship's capacity needed to be devoted to coal bunkers rather than cargo.

Engine efficiency was increased substantially when Elder and Randolph patented their compound engine in 1853, although its effective use was delayed until the introduction of higher-pressure boilers in the following decade. Graham (1956) documents the dramatic reduction in coal consumption during the second half of the nineteenth century: coal consumption per horsepower per hour of the average British steamship declined by more than half between 1855 and 1870 and it stabilized afterward (see Figure A.4 in the online Appendix). These improvements, in conjunction with the increase in the number of bunkering deposits, made steamship technology competitive even in long-distance trade. The transition was rapid. Figure 3 presents an aggregate representation of the transition from sail to steam. In 1869, the tonnage of British steam vessels engaged in international trade cleared in English ports surpassed that of British sailing vessels for the first time. Moreover,

TABLE 1—DESCRIPTIVE STATISTICS

	Mean	Median	SD	Min	Max	Observations
<i>Panel A. Unit of observation: country pair</i>						
Optimized shipping time—steam—Suez closed	434.55	397.50	250.54	0.00	1,054.48	8,834
Optimized shipping time—steam—Suez open	391.85	370.59	214.43	0.00	941.17	8,834
Optimized shipping time—sail—Suez open	914.81	857.18	513.98	0.00	2,418.92	8,834
Optimized shipping time—sail—Suez closed	855.28	818.78	461.76	0.00	2,166.01	8,834
Great-circle distance (km)	8,608.95	8,395.63	4,785.40	0.00	20,007.44	8,834
<i>Panel B. Unit of observation: port pair-year</i>						
Freight rates (shillings per ton)	17.11	15.38	10.81	2.49	390.00	4,903
Voyage duration (hours)	1,302.86	1,104.00	939.18	48.00	4,272.00	3,026
<i>Panel C. Unit of observation: country pair-year</i>						
Bilateral exports ('000 £)	2.29	0.45	6.00	0.00	138.80	23,885
<i>Panel D. Unit of observation: country-year</i>						
Total exports (1990-\$)	2.26e+09	7.36e+08	3.74e+09	2.70e+07	2.15e+10	344
Share nonagricultural exports	0.61	0.65	0.26	0.01	1.00	154
Per capita income (1990—intern.\$)	1,751.99	1,547.23	1,022.24	411.24	4,850.00	344
Total population ('000)	20,987.48	4,214.00	51,781.21	70.00	401,108.00	344
Urbanization rate (city size > 50,000)	0.10	0.08	0.09	0.00	0.38	344
Urbanization rate (city size > 100,000)	0.08	0.06	0.08	0.00	0.36	344
<i>Panel E. Unit of observation: country</i>						
Constraints on the executive (1860)	3.49	3.00	2.38	1.00	7.00	37
Colony (1850)	0.24	0.00	0.43	0.00	1.00	37

whereas sail powered more than two-thirds of the tonnage of ships built in the 1860s, this percentage declined to 15 percent during the early 1870s.

By the end of the 1880s, sailing vessels were still in use only in round-the-world trade, in Australian trade, and in trade to the west coast of the Americas. Finally, by 1910, the shift from sailing vessels to steamships was complete, and sailing vessels ceased to be used on a large scale in international trade.

II. Data

The aim of this article is to study (i) the impact of the adoption of the steamship on freight rates, patterns of international trade, and trade volumes throughout the world in the second half of the nineteenth century; (ii) the impact of the increase in world trade, resulting from the adoption of this new technology, on economic development; and (iii) the role of institutions and sectoral specialization in order to take full advantage of trade. To do this, a wealth of data are needed that are discussed in this section. In particular, data on shipping times by sail and by steam are described in Section IIA; data on freight rates in Section IIB; data on bilateral, sectoral and total trade in Section IIC; data on economic development, population, and urbanization in Section IID; and data on institutions in Section IIE. Table 1 reports the summary statistics for this set of data.

A. Sailing Times: Optimized Routes and Actual Voyages

Optimized bilateral sailing times were calculated by the author. The world was divided into a matrix of one-by-one degree squares. For each square, data downloaded from the Center for International Earth Science Information Network (CIESIN)³ were used to identify whether it was land or sea, while the US National Oceanic and Atmospheric Administration (NOAA) provided data on the average velocity and direction of the sea-surface winds.⁴

The sailing time from each oceanic square to each of the eight adjacent squares on the grid was determined by the velocity and direction of the wind along the path, according to the specific polar diagram of the vessel. The world matrix was then transformed into a weighted, directed graph in which every one-degree square is a node and the travel times to adjacent squares are the edges' weights.

Four graphs were constructed to account for the two sailing technologies (sail versus steam vessels) and the inclusion/exclusion of the Suez Canal as a valid path. Given any two nodes in the graph, the Dijkstra's algorithm was then used to compute the shortest travel time.⁵

After identifying the primary ports for each country, I calculated all pairwise minimum travel times. Identifying the primary ports for each country was straightforward, and for the majority of countries, the choice of port would not change the results. The exceptions were countries with the longest coastlines and those bordering two or more oceans. For these countries, up to five primary ports in 1850 were considered. The minimum travel time between two countries was then computed as the minimum travel time across all the ports of both countries.

The optimized routes by sail were compared with a set of actual voyages by sailing ships between 1742 and 1854 described in 3,026 logbooks digitized in the CLIWOC dataset.⁶ For these voyages, the logbooks report the starting point, the ending point, and the duration in days (intermediate stops are not recorded).⁷ Column 1 of Table 2 shows the results when regressing the duration of these voyages on the travel times computed using the optimization algorithm. The coefficient is positive, statistically significant (with *t*-statistics of approximately 16), and above 1, reflecting the fact that the optimized routes are direct routes, while actual routes might include intermediate stops (it should be noted that the *R*-square of this regression is approximately 50 percent). In column 2, I add year fixed effects: the results are unchanged. The partial scatter plot for the regression in this column is presented in Figure A.3 in the online Appendix. The results are also unchanged when controlling

³Data source: <http://sedac.ciesin.columbia.edu/data/collection/povmap>.

⁴Data source: https://www.nodc.noaa.gov/woce/woce_v3/wocedata_2/index.htm. Average monthly data are available from August 1999 to June 2002. Throughout the analysis I use averages for the month of May.

⁵Dijkstra's algorithm solves the single-source shortest path for arbitrary directed graphs with nonnegative weights from node *S* to node *E*. It is asymptotically the fastest known algorithm to solve this problem.

The algorithm starts at the initial node, *S*, and it grows a tree that ultimately spans all vertices reachable from *S*. Nodes are added to the tree in order of distance (i.e., first *S*, then the nodes closest to *S*, then the next closest, and so on). Every time a new node, *J*, is added to the tree, the algorithm computes the distance of all its neighbors to the node *J* and to the initial node *S*. The algorithm finished when eventually the tree reaches the end node *E*.

Specifically, the Dijkstra's algorithm is implemented in the Matlab module Dijkstra's Minimum Cost Path Algorithm: <http://www.mathworks.com/matlabcentral/fileexchange/20025-dijkstra-s-minimum-cost-path-algorithm>.

⁶Data source: <http://projects.knmi.nl/cliwoc/>.

⁷Voyages lasting less than a day or more than 180 days were excluded from the analysis.

TABLE 2—ACTUAL AND PREDICTED DURATION OF VOYAGES BY SAIL, 1742–1854

	Duration actual voyages (hours)			
	(1)	(2)	(3)	(4)
Sail distance (hours)	1.440 (0.0862)	1.392 (0.108)	1.158 (0.187)	1.115 (0.221)
Geographic distance			0.0423 (0.0209)	0.0406 (0.0231)
Intercept	432.6 (57.43)	1,136.2 (215.7)	331.1 (70.99)	1,335.0 (279.1)
Year dummies	No	Yes	No	Yes
R^2	0.485	0.529	0.494	0.536
Observations	3,026	3,026	3,026	3,026

Notes: The table reports OLS estimates. The unit of observation is the voyage in the CLIWOC dataset. The dependent variable is the actual duration of the voyage. Sail distance is the predicted duration of the voyage by sail computed using the optimization algorithm illustrated in Section IIA. Voyages lasting less than a day or more than 180 days are excluded from the sample. Standard errors (reported in parentheses) are three-way clustered (port of origin, port of destination, and year).

for geographic distance across starting and ending port (columns 3 and 4). Finally, Figure 5 shows the optimized routes by sailing vessels between England, Cape of Good Hope, and Java. The accuracy of the optimization is confirmed by the fact that these optimal routes can perfectly reproduce the routes followed by the actual journeys of British sailing ships shown in Figure 4, both in the Atlantic Ocean and in the Indian Ocean.⁸

Table 1, panels A and B, reports the summary statistics for this set of optimized shipping times and the duration of the actual voyages in the CLIWOC dataset. It is noteworthy that the introduction of the steamship reduced the average shipping time by more than half, and the opening of the Suez Canal reduced this time by an additional 10 percent.

B. Freight Rates

A database on freight rates from the English ports of Newcastle and Cardiff, covering the years 1855–1900, was constructed using three different sources.

The first source is the Newcastle Courant. This newspaper reported the freight rates for shipping coal from Newcastle to 147 different ports weekly. I collected the shipping rates reported in the first, second, and third week of February in the years 1855–1870. These freight rates were then averaged within each year-route, resulting in a total of 1,643 observations.

The second source is the Mitchell's Maritime Register, a weekly journal of shipping and commerce. This newspaper reported freight rates for shipping coal and iron

⁸ Note that I could not directly use the sailing time of these voyages in the empirical analysis because (i) they were available for only a very small subset of country pairs (less than 5 percent), and (ii) the stopping ports in these voyages were dictated not only by geography, but also by the map of economic development, which would have led to important endogeneity issues.

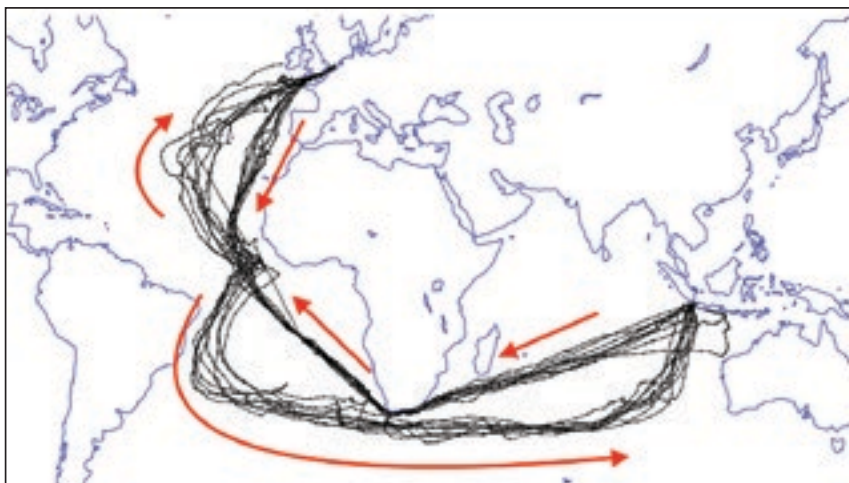


FIGURE 4. SHIPPING ROUTES BY SAILING SHIPS

Notes: The figure depicts 15 journeys made by British ships between 1800 and 1860. These journeys were randomly selected from the CLIWOC dataset among all voyages between England and Java comprised in the dataset.

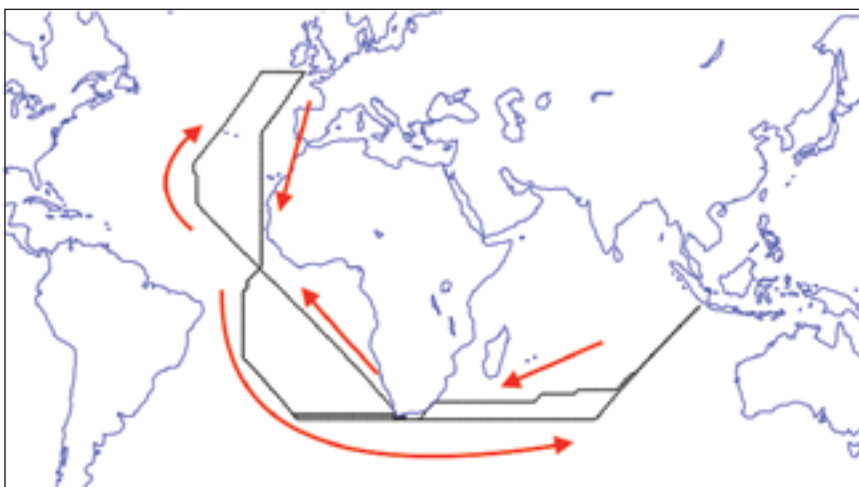


FIGURE 5. OPTIMAL ROUTES FOR SAILING VESSELS

Notes: The figure depicts the optimized routes by Clipper between England, Cape of Good Hope, and Java in the month of January.

in the years 1857–1883 from the ports of Cardiff and Newcastle. The entries from this source cover 226 routes for a total of 2,250 observations.

The last source is Angier (1920), which provides 1,010 observations of freight rates for coal and iron from the ports of Cardiff and Newcastle from the years 1870–1913.

The entries from these three sources were then standardized and converted into shillings per ton. Descriptive statistics are reported in Table 1, panel B.

C. Trade Data

A database for bilateral trade covering the entire second half of the nineteenth century was constructed by the author from primary sources. Table 1, panel C, reports the summary statistics for this set of data. Overall, the data consist of almost 24,000 bilateral trade observations for nearly 1,000 distinct country pairs. This database significantly improves upon the trade data used in prior studies of the nineteenth century, as it is better suited to identifying the impact of the steamship on trade patterns and development, mainly because of its sheer size and time coverage. To date, the most comprehensive bilateral trade database for this century is that constructed by Mitchener and Weidenmier (2008),⁹ which covers 700 distinct country pairs for the 1870–1900 period.¹⁰ My data are superior in both dimensions of the panel: the number of years and the country pairs. The most significant difference is that my data cover the entire second half of the century, which is essential to capturing the transition from sailing technology to steam. The list of countries with available bilateral trade data are listed in Table A.2 in the online Appendix.

A large number of documents, listed in the online Appendix, were used to assemble this dataset. Trade data for this period are available from summary tables assembled by the different national statistical institutes starting from national custom records. The great majority of the data (approximately 70 percent of the total entries) comes from the British Board of Trade Statistics. In particular, I rely on four different annual publications published by this institution between 1850 and 1905: the *Statistical Abstract for the Principal Foreign Countries*; the *Statistical Tables relating to Foreign Countries*; the *Statistical Abstract for the Several British Colonies, Possessions, and Protectorates*; and the *Statistical Abstract for the United Kingdom*. The second largest share of entries (11 percent) comes from the French Department of Foreign Affairs and Trade, and particularly from two annual publications: The *Tableau général du commerce de la France avec ses colonies et les puissances étrangères* and the *Mouvement général du commerce et la navigation des principaux pays étrangers*. Economic historians rank British and French trade statistics as the best data available for the nineteenth century. A complete discussion about their reliability can be found in Lampe (2008).¹¹ The remaining share of the entries comes from datasets assembled by contemporary historians on single (e.g., Pamuk 2010) or multiple countries (e.g., Mitchell 2007) based on national customs data.

The trade dataset also comprises 344 entries on total exports and 154 entries on the share of exports in nonagricultural products (37 countries every 5 years from

⁹I would like to thank Kris Mitchener for sharing his data with me.

¹⁰Other datasets on bilateral trade have been used in the literature. See Barbieri (1996); Lopez-Cordova and Meissner (2003); and Flandreau and Maurel (2001). All of these datasets begin after 1870, cover a much smaller number of dyads, and are overwhelmingly drawn from intra-European trade during the nineteenth century. Lampe (2008) provides bilateral trade flows in the period from 1857–1875, but only for Europe.

¹¹In particular, Lampe (2008, p. 7) writes “When overall data quality is referred to, the foreign trade statistics of the United Kingdom generally ranked as best practice [...] The literature on the quality of French trade statistics is small, but there are no hints that French statistics contained systematic errors, except for doubts on the accuracy of export price. Publications were extremely detailed and the classification did not change over time.”

1845 to 1905, with gaps). In order to construct the latter share, an approximate half million entries on exports by product were collected from primary sources; a SIC (rev1) code was then assigned to each of these products; and, finally, the share of exports that did not belong to the SIC categories 0 (food and live animals), 1 (beverages and tobacco), 4 (animal and vegetables oils and fat) was computed. Descriptive statistics for the variable total exports and share in nonagricultural products are reported in the first two rows of Table 1, panel D.

All trade data were then converted into pounds sterling using annual exchange rates provided by the British Board of Trade in numerous volumes of the *Statistical Abstract for the Principal and Other Foreign Countries* or by the Global Financial Database and Ferguson and Schularick (2006).

D. Per Capita GDP, Population, and Urbanization Rates

Data on per capita income were obtained from the Maddison Project Database (Bolt and van Zanden 2014), which is a recently updated version of the original Maddison (2004) dataset, whereas the population data come from a large number of different sources that are listed in the online Appendix.

This study also uses two different measures of urbanization: the percentage of the population living in cities with more than 50,000 and 100,000 citizens. Urbanization rate data were readily available for the majority of countries from the Cross-National Time-Series Data Archive (Banks and Wilson 2013). For the remaining ten countries in the sample, city-level data on the number of residents were obtained from a large number of sources (which are described in the online Appendix) and then aggregated at the country level.

The last four rows of Table 1, panel D, report summary statistics on per capita income, total population and urban population. The data are available every 5 years for the period from 1845–1905 for 37 countries (see Table A.3 in the online Appendix for the list of countries with available aggregate data).

E. Institutions

An initial question concerns which aspect of political institutions should be the focus of the analysis. North (1981) argues that high-quality institutions are a primary determinant of economic performance because they serve two functions: supporting private contracts (contracting institutions) and providing checks against expropriation by the government or other politically powerful groups (property rights institutions). However, Acemoglu and Johnson (2005), in an attempt to determine the relative roles of contracting institutions versus property rights institutions, find that only the latter have a first-order effect on long-term economic growth. For this reason, this paper will focus on the quality of property rights institutions and use the variable “constraints on the executive,” as defined in the dataset POLITY IV, to rank political institutions. This variable is designed to capture institutionalized constraints on the decision-making powers of chief executives. According to this criterion, better political institutions exhibit one or both of the following features: the holder of executive power is accountable to bodies of political representatives or to citizens, and/or government authority is constrained by checks and balances

and by the rule of law. A potential disadvantage of this measure is that it primarily concerns constraints on the executive while ignoring constraints on expropriation by other elites, including the legislature. The variable Constraints on the Executive varies from one (unlimited authority) to seven (accountable executive constrained by checks and balances) with higher values corresponding to better institutions. It is not available in the POLITY IV dataset for eight countries that were not independent in 1860. For these countries (highlighted in Table A.3 in the online Appendix), the score is therefore coded by the author.

III. The Steamships and The Effects on Trade

A. *The Shift from Sail to Steam*

The historical literature on when the introduction of steam technology to maritime transportation became relevant for international trade is divided. Graham (1956) and Walton (1970) argue that the transition from sail to steam was a slow and protracted process and was the result of the continuous improvements in the fuel consumption of marine engines that occurred throughout the second half of the century. By contrast, Fletcher (1958) and Knauerhase (1968) argue that the transition occurred fairly suddenly in the 1870s. In particular, Knauerhase attributes this change to the introduction of the compound engine, whereas Fletcher posits that it was the catalytic effect of the construction of the Suez Canal in 1860, which was suitable for steam vessels but not for sailing vessels.

Rather than assuming a particular position in this debate, I will use a gravity-type regression to determine when the distances in terms of the time to sail by steamship became relevant in explaining patterns of trade worldwide. The gravity model is an empirical workhorse in the trade literature. Practically, trade between two countries is inversely related to the distance between them and positively related to their economic size. The following is a basic expression for bilateral trade:

$$(1) \quad \ln(\text{trade}_{ijt}) = \ln(y_{it}) + \ln(y_{jt}) + \ln(y_{wt}) + (1 - \sigma) \ln(\tau_{ijt} + \ln P_{it} + \ln P_{jt}) + \varepsilon_{ijt},$$

where trade_{ijt} denotes the exports from country i to country j ; y_{it} and y_{wt} are the GDP of country i and of the world; τ_{ijt} is the bilateral resistance term (and captures all pair-specific trade barriers such as trade distance, common language, shared border, and colonial ties); and P_{it} and P_{jt} are the country-specific multilateral resistance terms that are intended to capture a weighted average of the trade barriers of a given country.

This specification emerges from several micro-founded trade models (see, for instance, Anderson and van Wincoop 2003 and Eaton and Kortum 2002). These models typically imply a set of predictions regarding trade diversion and trade creation. First, exports from i to j are increased when the bilateral resistance term τ_{ijt} declines relative to the multilateral resistance terms P_{it} and P_{jt} . Second, as world trade is homogeneous of degree zero in the bilateral resistance terms, international trade will increase only when international frictions τ_{ijt} and τ_{jit} decline relative to intranational frictions τ_{iit} and τ_{jjt} . Note that the introduction of the steamship was responsible for both a change in the relative bilateral frictions across countries and

a reduction in international frictions relative to intranational frictions, as the steamship was utilized disproportionately more for international shipping than for domestic shipping.

Although the majority of international trade is shipped by sea, the vast majority of estimated gravity models assume that the bilateral resistance term is a function of point-to-point great circle distances rather than navigation distances. By contrast, this paper assumes that this term is a function of shipping times by both sail and steam vessels. In particular, I will estimate the following equation:

$$(2) \quad \ln(\text{trade}_{ijt}) = \beta_{\text{steam},T} \ln(\text{steamTIME}_{ij}) + \beta_{\text{sail},T} \ln(\text{sailTIME}_{ij}) \\ + X_{ijt} \Gamma + \gamma_t + \varepsilon_{ijt},$$

where steamTIME_{ij} and sailTIME_{ij} are the sailing times from country i to country j by steam and sailing vessels, respectively, and X_{ijt} indexes a set of variables to control for the P and y terms in the original gravity equation. Note that the coefficients on the two distances are allowed to vary over time to capture changes in the navigation technology from sail to steam.

The results of these regressions are presented in Table 3. Standard errors are three-way clustered to allow for arbitrary correlation within exporter, importer, and year. The coefficient on sailing times is allowed to vary every five years. In the benchmark specification, the P and y terms are controlled using importer fixed effects, exporter fixed effects, and year fixed effects. Figure 6 plots the sequence of the estimated coefficients on shipping times (the error bars represent the 90 percent confidence interval around the point estimates). The coefficient on shipping time by steam is close to zero and not significant until 1865, while it becomes large and negative thereafter: it decreases to -0.4 in the period from 1866–1870, to -0.6 in the period from 1871–1875, and it stays at similar levels until 1900. In the same figure, the coefficient on shipping time by sail is negative (between -0.8 and -0.7) until 1865, it decreases to -0.3 in the period from 1866–1870, while it moves toward zero in the years thereafter. This evidence is consistent with the view of a rapid change toward steam in the maritime transportation industry that started in the late 1860s and was completed in the late 1870s.¹²

A potential concern with this specification is that countries' relative sizes and multilateral resistance change over time. If these relative changes are correlated with shipping distances by sail and steam, then the estimates in Figure 6 may be biased. For this reason, I supplement this specification with importer-by-year fixed effects and exporter-by-year fixed effects. Figure 7 indicates that from a qualitative perspective, the results are only slightly affected. In a practical sense, the only difference is that in this new specification, the coefficient on shipping times by steam

¹²In Table A.4 in the online Appendix, I split the sample by voyage length (above and below the mean) and estimate equation (2) for the two subsamples. Also in this case, the regression includes year and pair fixed effects. The estimates are generally more imprecise. However, it is important to note that the bulk of the variation in patterns of trade due to the steamship happens between 1866 and 1875 independent of whether we examine short routes or long routes. In both cases, the estimated coefficient on $\ln(\text{sailTIME}_{ij})$ in the years 1861–1865 is small and not significant, it increases substantially in the years 1866–1870 and 1871–1875, and it stabilizes for the following years. The estimates on the coefficients on $\ln(\text{steamTIME}_{ij})$ are more noisy but, from a qualitative point of view, exhibit an opposite pattern and are not substantially different between shorter and longer routes.

TABLE 3—THE SHIFT FROM SAIL TO STEAM: SHIPPING TIMES AND EXPORTS

	log (export/population)				
	(1)	(2)	(3)	(4)	(5)
$\ln(\text{steam dist}) \times I(\text{year} \leq 1860)$	0.390 (0.421)	0.0712 (0.439)	0.978 (0.658)	0.661 (0.687)	—
$\ln(\text{steam dist}) \times I(1861\text{--}1865)$	0.222 (0.551)	−0.104 (0.553)	0.438 (0.681)	0.102 (0.716)	−0.0994 (0.231)
$\ln(\text{steam dist}) \times I(1866\text{--}1870)$	−0.392 (0.328)	−0.714 (0.361)	−0.904 (0.302)	−1.197 (0.383)	−0.270 (0.200)
$\ln(\text{steam dist}) \times I(1871\text{--}1875)$	−0.604 (0.278)	−0.926 (0.330)	−0.677 (0.300)	−0.988 (0.372)	−0.496 (0.221)
$\ln(\text{steam dist}) \times I(1876\text{--}1880)$	−0.682 (0.268)	−1.004 (0.324)	−0.628 (0.295)	−0.940 (0.367)	−0.577 (0.233)
$\ln(\text{steam dist}) \times I(1881\text{--}1885)$	−0.594 (0.300)	−0.927 (0.367)	−0.543 (0.279)	−0.868 (0.321)	−0.491 (0.227)
$\ln(\text{steam dist}) \times I(1886\text{--}1890)$	−0.644 (0.311)	−0.979 (0.391)	−0.570 (0.282)	−0.898 (0.342)	−0.545 (0.224)
$\ln(\text{steam dist}) \times I(1891\text{--}1895)$	−0.675 (0.287)	−1.008 (0.393)	−0.560 (0.251)	−0.889 (0.380)	−0.603 (0.203)
$\ln(\text{steam dist}) \times I(1896\text{--}1900)$	−0.578 (0.262)	−0.912 (0.354)	−0.601 (0.246)	−0.931 (0.360)	−0.534 (0.210)
$\ln(\text{sail dist}) \times I(\text{year} \leq 1860)$	−0.861 (0.395)	−0.784 (0.392)	−1.344 (0.674)	−1.265 (0.674)	—
$\ln(\text{sail dist}) \times I(1861\text{--}1865)$	−0.720 (0.550)	−0.634 (0.542)	−0.891 (0.690)	−0.797 (0.701)	0.141 (0.267)
$\ln(\text{sail dist}) \times I(1866\text{--}1870)$	−0.327 (0.314)	−0.247 (0.317)	0.117 (0.326)	0.174 (0.339)	0.266 (0.211)
$\ln(\text{sail dist}) \times I(1871\text{--}1875)$	−0.193 (0.277)	−0.122 (0.279)	−0.156 (0.319)	−0.0923 (0.325)	0.456 (0.217)
$\ln(\text{sail dist}) \times I(1876\text{--}1880)$	−0.0958 (0.270)	−0.0249 (0.270)	−0.192 (0.313)	−0.126 (0.320)	0.556 (0.226)
$\ln(\text{sail dist}) \times I(1881\text{--}1885)$	−0.118 (0.314)	−0.0347 (0.318)	−0.169 (0.290)	−0.0878 (0.284)	0.540 (0.231)
$\ln(\text{sail dist}) \times I(1886\text{--}1890)$	−0.105 (0.315)	−0.0192 (0.323)	−0.139 (0.284)	−0.0505 (0.282)	0.599 (0.217)
$\ln(\text{sail dist}) \times I(1891\text{--}1895)$	−0.0321 (0.289)	0.0527 (0.305)	−0.131 (0.252)	−0.0399 (0.281)	0.713 (0.194)
$\ln(\text{sail dist}) \times I(1896\text{--}1900)$	−0.0873 (0.271)	−0.00152 (0.280)	−0.108 (0.249)	−0.0153 (0.267)	0.715 (0.185)
$\ln(\text{geo dist})$		0.269 (0.192)		0.259 (0.190)	
Importer fixed effects	Yes	Yes	No	No	No
Exporter fixed effects	Yes	Yes	No	No	No
Year fixed effects	Yes	Yes	No	No	Yes
Pair fixed effects	No	No	No	No	Yes
Importer $\times$ year fixed effects	No	No	Yes	Yes	No
Exporter $\times$ year fixed effects	No	No	Yes	Yes	No
R^2	0.632	0.633	0.169	0.171	0.803
Observations	23,885	23,885	23,885	23,885	23,885

Notes: The table reports OLS estimates. The unit of observation is a country pair in a certain year. The dependent variable is the log export-to-population ratio. Standard errors (reported in parentheses) are three-way clustered (exporter, importer, and year).

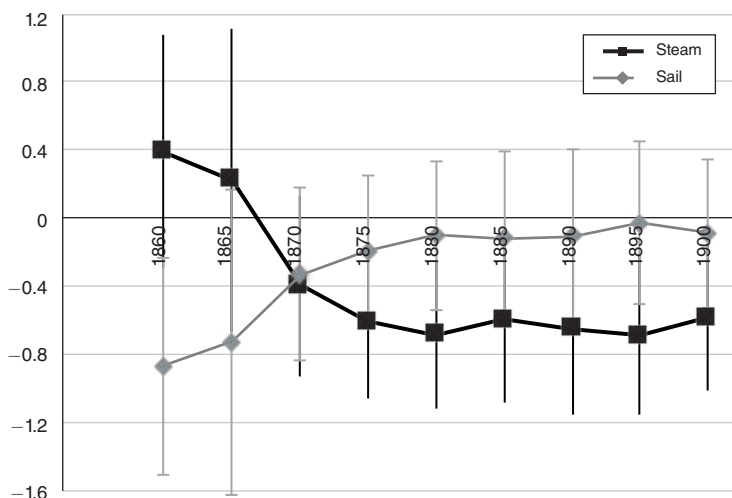


FIGURE 6. THE CHANGE IN THE ELASTICITY OF TRADE WITH RESPECT TO SHIPPING TIMES BY SAIL AND STEAM VESSELS (*Estimates from a Gravity Model with Importer, Exporter, and Year Fixed Effects*)

is positive and large in the years before 1860, although it is not significant at conventional statistical levels.

Finally, the coefficients plotted in Figure 8 come from a regression that includes year and bilateral pair dummies. In this case, the absolute level of the elasticities of sailing times cannot be captured; instead, it is only possible to observe their change over time. For this reason, the level in 1860 is standardized to 0. These estimates match the previous findings. Over time, country pairs that were relatively closer by steam than by sail experienced the greatest increase in trade.¹³

Overall, these results corroborate the view that the introduction of the steamship in the shipping industry was responsible for a substantial change in trade patterns in the second half of the nineteenth century, with the majority of the change happening during the late 1860s and the 1870s.

B. The Steamship and the First Wave of Globalization

The previous section emphasized that the steamship reshaped global trade patterns beginning in approximately 1870. Circa this period, per capita international trade increased three-fold. Is there a causal link between these two observations? What extent of the increase in international trade is explained by the shift from sail to steam in the maritime industry? Estevadeordal, Frantz, and Taylor (2003) argue that a general equilibrium gravity model of international trade implies that 57 percent of the world trade boom between 1870 and 1913 can be explained by income growth. The adoption of several currency unions and declining freight rates

¹³Controlling for great circle distances across pairs does not alter the results (see column 2 in Table 3). It is noteworthy that, once controlling for shipping times, geographic distance no longer exerts a negative effect on bilateral trade.

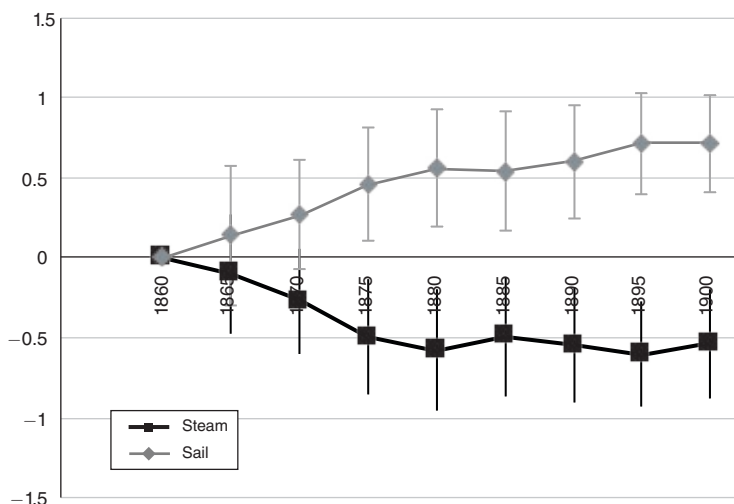


FIGURE 7. THE CHANGE IN THE ELASTICITY OF TRADE WITH RESPECT TO SHIPPING TIMES BY SAIL AND STEAM VESSELS (Estimates from a Gravity Model with Importer $\times$ Year and Exporter $\times$ Year Fixed Effects)

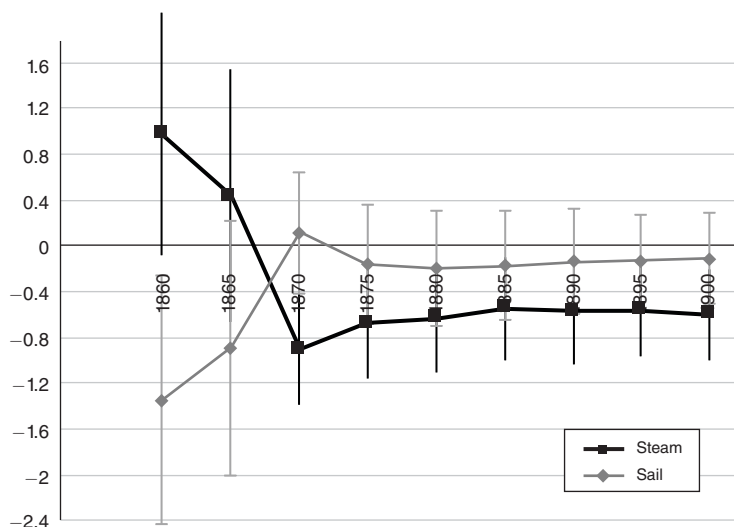


FIGURE 8. THE CHANGE IN THE ELASTICITY OF TRADE WITH RESPECT TO SHIPPING TIMES BY SAIL AND STEAM VESSELS (Estimates from a Gravity Model with Country-Pair and Year Fixed Effects)

each roughly account for one-third of the remaining part, while the remainder is explained by income convergence and tariff reductions. A more recent work by Jacks, Meissner, and Novy (2011) corrects the estimates obtained by Estevadeordal Frantz, and Taylor. (2003) using a more comprehensive dataset on freight rates for the same period and finds the effect of the maritime transport revolution on the late nineteenth century global trade boom to be trivial. Other studies focus on global market integration—the convergence of prices across markets—rather than on trade and trade shares. Jacks (2006) presents evidence from a number of North Atlantic grain markets between 1800 and 1913 and indicates that changes in freight costs

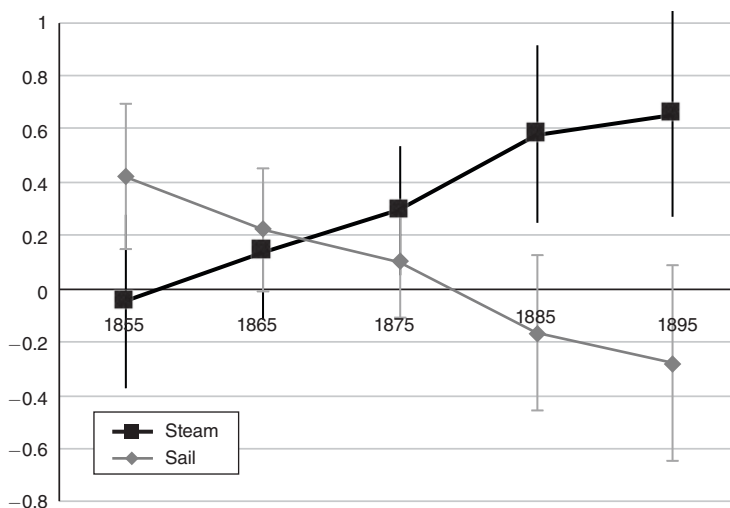


FIGURE 9. THE CHANGE IN THE ELASTICITY OF FREIGHT RATES WITH RESPECT TO SHIPPING TIMES BY SAIL AND STEAM VESSELS (*Estimates from a Gravity Model with Country of Destination, Product, and Year Fixed Effects*)

can explain only a relatively modest fraction of the changes in trade costs occurring in those markets. Using similar data, Federico and Persson (2007) conclude that changes in trade policies were the single most important factor explaining the convergence and divergence of prices in the long term.

All of the above-mentioned studies use changes in freight rates to proxy for changes in transportation costs. The disadvantage of this approach is that freight rates are simply prices for transport services and, as such, are likely to respond not only to technology shocks but also to shifts in the demand schedule for shipping services and changes in the market structure in the shipping industry. Both of these confounding factors were present in 1870. First, the adoption of the gold standard, income growth and more liberal trade policies may have generated an outward shift in the demand for shipping services. Second, beginning in the 1870s, rate schedules and shipping capacity for overseas freight on a number of trade routes began to be established by shipping line conferences/cartels. Every main trade route between regions had its own shipping conference organization composed of all the shipping lines that served the route.¹⁴

Given the contemporaneous shift in the demand for shipping services and the change in market structure, it is not surprising that the introduction of the steamship did not immediately translate into a sharp reduction in the overall price of shipping services. For instance, the North's freight index of American export routes declined at the beginning of the nineteenth century and remained stable between 1850 and

¹⁴ Morton (1997, p. 51) explains as follows: "The purpose of the shipping conference was to set rates and sailing schedules to which each line would adhere. The cartel also allocated market shares of specific types of goods and decided the exact ports to be served by each member line [...]. By the turn of the century, most shipping routes had been cartelized [...]." Morton (1997) notes that these shipping conferences were able to establish prices above marginal costs and to control shipments such that members were extracting monopolistic rents. Defying the cartel was difficult, and most conferences would share revenues. Finally, entry was generally prevented through price predation, although some entrants were formally admitted to the cartel without conflict (Podolny and Morton 1999).

1880, whereas Harley's British index declined more rapidly after 1850, well before the introduction of the steamship on a major trade route. Neither index exhibited a structural break between 1865 and 1880, although the number of steamers that were constructed increased significantly during this period, while the construction of larger sailing ships nearly ceased.¹⁵

In this section, I will measure the effect of the introduction of the steamship on the trade boom during the second half of the nineteenth century by relying on an actual measure of technological improvement—the reduction in sailing times—rather than on freight rate indexes. This approach has the key advantage that the change in sailing times is arguably exogenous with respect to the demand for shipping services and the market structure in the shipping industry, as this change is the result of pre-vailing wind patterns and ocean currents.

Table 4 reports estimates from the following regression:

$$(3) \quad \ln(\text{freight}_{ijpt}) = \beta_{\text{steam}, T} \ln(\text{steamTIME}_{ij}) + \beta_{\text{sail}, T} \ln(\text{sailTIME}_{ij}) \\ + X_{ijpt} \Gamma + \gamma_t + \varepsilon_{ijpt},$$

where freight_{ijpt} is the average freight rate from port i to port j for product p in year t ; X_{ijpt} indexes a set of control variables; and γ_t are year fixed effects that are supposed to capture common shocks to freight rates. As in equation (2), the coefficients on the two shipping times are allowed to vary over time to capture changes in the navigation technology from sail to steam. Unfortunately data on freight rates are not as comprehensive as data on exports: they are limited to outbound rates on 2 products—coal and iron—and only capture shipments from two UK ports—Cardiff and Newcastle—to 195 foreign ports (for a total of 291 routes) scattered over 33 different countries. To increase the power of the OLS, the coefficients on sailing times are allowed to vary every 10 years (rather than every 5 years, as for the trade estimates in Table 3).

The benchmark specification, reported in column 1 of Table 4, controls for country of destination, year, and product fixed effects. Voyage duration by steamship does not have a relevant impact on freight rates in the 1850s and the 1860s, while it does have a positive and significant impact thereafter. Voyage duration by sailing ship instead has a positive impact in the first two decades but not thereafter. These results are unchanged when controlling for great-circle distance between ports (column 2) and when adding route fixed effects (column 3), with the usual caveat that, in this latter case, the absolute level of elasticities of sailing times cannot be captured.

In sum, the results in Table 4 prove that, although there was not a general sharp decline in freight rates immediately after the diffusion of the steamship, there was still a relative decline along those routes in which the steamship had a larger impact on shipping times. To estimate the effect of the reduction in shipping times induced

¹⁵ For instance, in the Angier Brothers' freight report for 1871, we read the following: the number of new sailing vessels is unprecedentedly small, whereas the increase in the number of steamers is almost double that of any preceding years. See also Figure 3.

TABLE 4—THE SHIFT FROM SAIL TO STEAM: SHIPPING TIMES AND FREIGHT RATES

	log (freight rate)		
	(1)	(2)	(3)
$\ln(\text{steam dist}) \times I(1855\text{--}1860)$	−0.0492 (0.199)	−0.0128 (0.187)	—
$\ln(\text{steam dist}) \times I(1861\text{--}1870)$	0.135 (0.156)	0.173 (0.151)	0.105 (0.173)
$\ln(\text{steam dist}) \times I(1871\text{--}1880)$	0.287 (0.146)	0.331 (0.145)	0.170 (0.196)
$\ln(\text{steam dist}) \times I(1881\text{--}1890)$	0.575 (0.201)	0.620 (0.189)	0.426 (0.241)
$\ln(\text{steam dist}) \times I(1891\text{--}1900)$	0.652 (0.234)	0.705 (0.232)	0.494 (0.238)
$\ln(\text{sail dist}) \times I(1855\text{--}1860)$	0.420 (0.165)	0.403 (0.156)	—
$\ln(\text{sail dist}) \times I(1861\text{--}1870)$	0.225 (0.138)	0.206 (0.131)	−0.125 (0.144)
$\ln(\text{sail dist}) \times I(1871\text{--}1880)$	0.109 (0.129)	0.0853 (0.127)	−0.176 (0.163)
$\ln(\text{sail dist}) \times I(1881\text{--}1890)$	−0.164 (0.176)	−0.188 (0.168)	−0.415 (0.204)
$\ln(\text{sail dist}) \times I(1891\text{--}1900)$	−0.279 (0.222)	−0.311 (0.219)	−0.540 (0.205)
$\ln(\text{geo dist})$		−0.0407 (0.0973)	
Country of destination fixed effects	Yes	Yes	No
Year fixed effects	Yes	Yes	Yes
Route fixed effects	No	No	Yes
Product fixed effects	Yes	Yes	Yes
R^2	0.306	0.306	0.0294
Observations	4,903	4,903	4,903

Notes: The table reports OLS estimates. The unit of observation is a port pair in a certain year. The dependent variable is the log freight rate. Standard errors (reported in parentheses) are two-way clustered (country of destination and year).

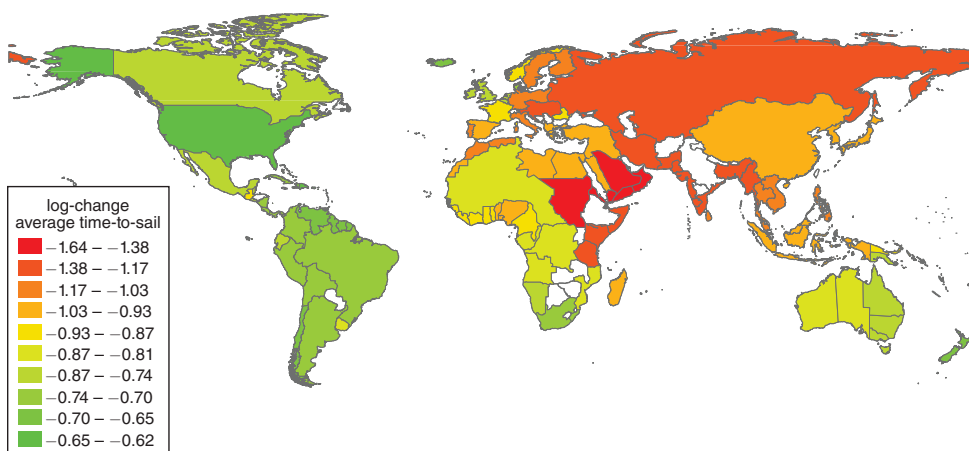
by the introduction of the steamship on the change in international trade volumes, I then estimate the following regression:

$$(4) \quad \Delta \log T_i = c + \alpha \Delta \log Dist_i + v_i,$$

where c is the intercept, $\Delta \log T_i$ is the log-change in either the export-to-GDP ratio or the per capita exports of country i between 1850 and 1905, and $\Delta \log Dist_i$ is the average change in shipping times across all trading partners (weighted by their share of world trade) generated by the introduction of the steamship:

$$(5) \quad \Delta \log Dist_i \equiv \sum_{j \neq i} w_j [\ln(\text{steam}TIME_{ij}) - \ln(\text{sail}TIME_{ij})].$$

Figure 10 reports the different values of the $\Delta \log Dist_i$ variable, across all the polities of the world in 1900 that are not landlocked. (See the online Appendix C for an in-depth discussion of the geographical determinants of this variable.)

FIGURE 10. LOG-CHANGE IN THE AVERAGE TIME-TO-SAIL (*From Sail to Steam*)

The elasticity α can be interpreted as the effect of the introduction of the steamship on international trade by reducing sailing time, under the assumption that all international trade was carried by sailing vessels in 1850 and by steam vessels in 1905. Because a smaller portion of international trade was still conducted by sail in 1905 or was shipped by land (or river), estimates of the effects of the steamship are likely to be downwardly biased.

The results are reported in Table 5 (but also in Figures A.5 and A.6 in the online Appendix). The benchmark specification, reported in column 1, implies that the effect of isolation on trade is negative and large. Increasing the weighted average shipping time to the rest of the world from the level of France to the level of Cuba implies a reduction in the export-to-GDP ratio of 141 log points and in the export-to-population ratio of 243 log points. In columns 2, 3, 5, and 6 of Table 5, I limit the weighted sailing distances to the United Kingdom and top 5 trading countries. It is clear that the qualitative results do not vary according to the particular weights selected to aggregate sailing times across the different trading partners. In all cases, the effect of isolation is negative, although the estimated elasticity oscillates between -0.8 and -1.5 when the regressor is the export-to-GDP ratio and between -1.4 and -2.5 when the regressor is per capita exports. It should be noted that there are more observations for the latter variable (50 versus 25) because we lack GDP data for the years 1850 and 1905 for a large number of countries. The result is that the OLS estimates are more precise when using per capita exports rather than the export-to-GDP ratio. In Table 5, the observations are weighted by the log of the countries' total populations. Unweighted estimates are reported in the online Appendix: the results are practically unchanged (see Table A.5).

These estimated elasticities can be used to produce a rough estimate of the role of the introduction of steam vessels in spurring trade during the period of analysis. The population-weighted average log-change in per capita trade between 1850 and 1905 in my sample of countries is 1.4. If we assume that the steamship in 1905 is, on

TABLE 5—GEOGRAPHICAL ISOLATION AND TRADE

	log change export/GDP			log change export/population		
	(1)	(2)	(3)	(4)	(5)	(6)
log-change distance (to all countries)	−1.008 (0.663)			−1.737 (0.480)		
log-change distance (to United Kingdom)		−0.792 (0.469)			−1.422 (0.341)	
log-change distance (to top five trading countries)			−1.498 (0.874)			−2.469 (0.662)
Intercept	−0.177 (0.582)	−0.00547 (0.436)	−0.0416 (0.441)	−0.389 (0.419)	−0.141 (0.308)	−0.100 (0.331)
R^2	0.0912	0.115	0.113	0.214	0.270	0.225
Observations	25	24	25	50	49	50
Weighted (by log population)	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports OLS estimates. The unit of observation is the country. The dependent variable is the log-change of either export/GDP or export/population of the country between 1850 and 1905. log-change distance is the weighted average of the log changes in shipping times between the country and the other countries of the world generated by the introduction of the steamship (see equation (5) in the main text). Robust standard errors are reported in parentheses.

average, 50 log-points faster than the sailing vessels active in 1850¹⁶ then the most conservative estimates imply that the steamship might be responsible for at least half ($-0.5 \times -1.44/1.4 = 0.51$) of the trade boom that occurred over these years.¹⁷ This number is surprisingly large compared with the previous estimates described at the beginning of this section. However, as usual in these back-of-the-envelope calculations, we should take it with a grain of salt, as it is based on the assumption that the rollout of the steam was uniform across different countries and different products and it was completely concluded by the end of the period of analysis.

To conclude, starting in 1865–1870, the introduction of the steamship reduced shipping times by approximately one-half. This did not translate in an immediate reduction of average freight rates following the introduction of the steamship because other confounding factors were, at the same time, pushing freight rates in

¹⁶To compute the average reduction in shipping times induced by the steamship, I use a novel database based on the Atlantic Canada Shipping Project, which was a major project undertaken by the Canadian Maritime History Group. This database contains information on all vessels registered in the ten major Atlantic Canadian ports between 1787 and 1936 and on approximately 16,000 voyages that were undertaken by these vessels in the years 1864–1914. There are 70 routes in which both sailing and steam vessels operated (data on voyage duration on these routes are available for 1,667 voyages).

I then regress the log duration of these voyages on a dummy that identifies those carried by steam vessels plus directed route fixed effects and, depending on the specification, year fixed effects (see Table A.6 in the online Appendix).

In the specification without year fixed effects, the log-difference in the duration of voyages between sail and steam along the same route is 0.489 (*t*-statistics 2.18). When adding year fixed effects, the results are similar, although the standard errors become larger: this result is not surprising because this specification is particularly demanding given that the majority of sailing voyages are concentrated at the beginning of the period while the majority of steam voyages are at its end). Additionally, the coefficient on the steam dummy is smaller (from 0.489 to 0.408) because year fixed effects absorb the increase in the relative speed of the steam vessels with respect to the sailing vessels in the time period under analysis.

¹⁷Note that the estimates from equation (4) are preferable to the estimates from equation (2) to serve as a basis to evaluate the impact of the steamship on trade. They are robust to potential externalities to the rest of the world, in terms of trade diversion or trade creation, that might arise from an exogenous increase in bilateral trade between any pair of countries.

the opposite direction. A simple back-of-the-envelope calculation (which should be taken cautiously) suggests that, the fact that the steamship was able keep freight rates on the low side might be the major determinant of the first wave of trade globalization.

IV. Trade and Economic Development

The aim of this section is to evaluate the effect of this trade boom in the second half of the nineteenth century on economic development. The basic estimating equation is as follows:

$$(6) \quad \log(Y_{it}) = \gamma \log T_{it} + \gamma_i + \gamma_t + v_{it},$$

where Y_{it} is per capita GDP, and T_{it} is either the export-to-GDP ratio or the per capita exports of country i . To identify the causal effect, this equation is estimated using 2SLS, instrumenting T_{it} with the component of country i 's total exports that is explained by the geographic isolation of the country as determined by the prevailing shipping technology in t . Specifically, I isolate the geographic component of country i 's exports to its trade partner j in year t using the following formula:

$$(7) \quad \log PT_{ijt} = \hat{\beta}_{steam,t} \ln(steamTIME_{ij}) + \hat{\beta}_{sail,t} \ln(sailTIME_{ij}).$$

The geographic component of a country's total exports is then computed as the weighted average of these bilateral components across all of country i 's potential trading partners using the partners' shares in total world trade as weights:

$$(8) \quad \log PT_{it} = \sum_{j \neq i} w_j \log PT_{ijt}.$$

Note that the instrument for trade, $\log PT_{it}$, is time varying. Within-country variation is generated by the shift from sail to steam vessels, which induces a change in the bilateral shipping time across countries and, through this channel, a shift in the relative level of geographic isolation of countries worldwide. The time-varying nature of the instrument implies that, in contrast to the approach used by Frankel and Romer, country fixed effects can be added to equation (6).

Table 6 presents the OLS, 2SLS, and reduced-form estimates of equation (6). Standard errors are two-way clustered to allow arbitrary correlations within country and within year and they are corrected to account for the fact that the instrument depends on the (estimated) parameters of the bilateral trade equation.¹⁸ The sample is an unbalanced panel that covers 37 countries with observations every 5 years from 1845 to 1905 (the exact countries and years available are illustrated in

¹⁸The variance-covariance matrix is estimated as the usual IV formula for errors clustered at the country-year level plus $\left(\frac{\partial \hat{\gamma}}{\partial \hat{\beta}}\right) \hat{\Omega} \left(\frac{\partial \hat{\gamma}}{\partial \hat{\beta}}\right)'$, where $\hat{\gamma}$ is the vector of estimated coefficients from the income regression, $\hat{\beta}$ is the vector of estimated coefficients from the bilateral trade equation, and $\hat{\Omega}$ is the estimated variance-covariance matrix of $\hat{\gamma}$. (The computation of $\frac{\partial \hat{\gamma}}{\partial \hat{\beta}}$ has been done numerically as in Frankel and Romer 1999.)

TABLE 6—TRADE AND DEVELOPMENT

	log per capita GDP							
	(OLS) (1)	(2SLS) (2)	(2SLS) (3)	(OLS) (4)	(2SLS) (5)	(2SLS) (6)	(OLS) (7)	(OLS) (8)
<i>Panel A</i>								
log (export/GDP)	−0.054 (0.031)	−0.200 (0.087)	−0.187 (0.094)					
log (export/pop)				0.137 (0.038)	−0.250 (0.136)	−0.230 (0.142)		
log predict. export							−0.215 (0.109)	−0.194 (0.112)
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	344	344	344	344	344	344	344	344
<i>F</i>		26.50	24.98		16.77	16.56		
Weighted	No	No	Yes	No	No	Yes	No	Yes
<i>Panel B</i>								
log predict.		1.075	1.038		0.860	0.844		
Export		(0.387)	(0.377)		(0.340)	(0.334)		

Notes: The table reports OLS and 2SLS estimates. The unit of observation is country-year. The dependent variable is the log of per capita GDP. “log predict export” is constructed according to equation (8). Observations are un-weighted in columns 1, 2, 4, 5, and 7 and weighted by the log-population of the country in 1860 in the other columns. Panel A reports the second-stage estimates. *F* is the Kleiberg-Paap Wald *F*-statistics for weak identification. Panel B reports the first-stage estimates. Standard errors (reported in parentheses) are two-way clustered (country and year) corrected to account for the fact that the instrument depends on the (estimated) parameters of the bilateral trade equation.

Figure A.7 in the online Appendix). The OLS estimates (columns 1 and 4) produce opposite results depending on the measure of trade openness. Per capita GDP is negatively correlated with the export-to-GDP ratio and positively correlated with the export-to-population ratio. This anomaly is likely the result of a spurious correlation of the denominators of the two regressors with the dependent variable. Columns 2 and 5 (3 and 6) report the unweighted (weighted) 2SLS estimates. In each case, the first stage is strong. The Kleiberg-Paap *F*-statistic for weak identification exceeds 10, which is the standard threshold for a powerful instrument as suggested by Staiger and Stock (1997). The impact of trade openness on per capita GDP is negative. An increase in the export-to-GDP (export-to-population) ratio by 1 percent produces a reduction in per capita GDP in the order of 0.20 (0.25) percent. Finally columns 7 and 8 report the reduced-form estimates.

To understand which of the observations are driving the first stage and the reduced-form estimates, Figures A.8–A.10 in the online Appendix provide a simple scatter plot of the log-change in predicted trade, PT_{it} , against the log-change in export rates and in per capita GDP over the years 1850–1905.¹⁹ Figure A.8 shows that there is a clear positive correlation between the log-change in the instrument and the log-change in both the export-to-population and the export-to-GDP ratios. Figure A.9 reports, instead, the scatter plot of the log-change in the instrument

¹⁹ Note that all countries for which data are not available both in 1850 and 1905 are not included in these figures: we are left with 25 countries for the first stage and 29 for the reduced form (rather than 37 countries, as in Table 6).

TABLE 7—TRADE, POPULATION, AND URBANIZATION RATES

	log (Population)			log (Urban pop (>50,000))			log (Urban pop (>100,000))		
	(OLS) (1)	(2SLS) (2)	(2SLS) (3)	(OLS) (4)	(2SLS) (5)	(2SLS) (6)	(OLS) (7)	(2SLS) (8)	(2SLS) (9)
log (export/GDP)	-0.243 (0.074)	-1.041 (0.196)	-0.972 (0.177)	-0.013 (0.013)	-0.065 (0.035)	-0.061 (0.037)	-0.243 (0.074)	-0.069 (0.049)	-0.068 (0.049)
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	344	344	344	344	344	344	344	344	344
<i>F</i>		26.50	24.98		26.50	24.98		26.50	24.98
Weighted	No	No	Yes	No	No	Yes	No	No	Yes

Notes: The table reports OLS and 2SLS estimates. The unit of observation is country-year. The dependent variable is the log of population (columns 1–3) or the log of the population share living in cities with more than either 50,000 citizens (columns 4–6), or 100,000 citizens (columns 7–9). Observations are un-weighted in columns 1, 2, 4, 5, 7, and 8 and weighted by the log-population of the country in 1860 in columns 3, 6, and 9. *F* is the Kleiberg-Paap Wald *F*-statistics for weak identification. Panel B reports the first-stage estimates. Standard errors (reported in parentheses) are two-way clustered (country and year) corrected to account for the fact that the instrument depends on the (estimated) parameters of the bilateral trade equation.

against the log-change in per capita GDP: the two variables are negatively correlated. The main exceptions to this rule are four South American republics: Brazil, Venezuela, Colombia, and Uruguay. A potential explanation is that Latin America had the highest tariffs in the world from the late 1880s onward (Williamson 2011, p. 204). In these countries, shipping rates represented a small fraction of total trade costs, and the change in freight rates induced by the steamship was unlikely to have had a large impact on total trade costs. In Figure A.9, Europe is clearly dominating the picture, as 14 out of 29 countries are European. Moreover, European countries experienced the largest reduction in shipping times in this sample and faced relatively low per capita GDP growth rates (this is particularly true for Southern Europe). In Figure A.10, European countries are omitted: the correlation is still negative and the slope of the regression line is practically unchanged. In this case, however, the correlation coefficient drops by half when excluding New Zealand and Dutch East Indies from the sample.

A potential concern with these estimates is that per capita GDP might not be an ideal proxy of economic development in a world that is largely still Malthusian. For this reason, in Table 7, I repeat the analysis using population and urbanization rates as alternative proxies. Trade again has a negative impact on both variables. An increase in the export-to-GDP ratio by 1 percent decreases population by 1 percent, the share of the population living in cities with more than 50,000 citizens by 0.06 percent and the share living in cities with more than 100,000 citizens by 0.07 percent. The results are practically unchanged when using per capita trade as a measure of trade openness (see Table A.7 in the online Appendix).

The finding that the effect of the first wave of globalization could be negative on average is surprising. In a previous study, Williamson (2011) documents a negative correlation between growth in terms of trade (generated by increased trade) and per capita GDP growth in a large set of developing countries between 1870

TABLE 8—TRADE AND ECONOMIC DIVERGENCE

Dependent variable: log per capita GDP	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log export/GDP	−0.400 (0.180)	−0.395 (0.182)	−0.241 (0.0961)	−0.219 (0.0987)				
log export/population					−0.419 (0.217)	−0.407 (0.219)	−0.260 (0.122)	−0.227 (0.122)
log export/GDP × above mean GDP 1850	0.248 (0.143)	0.254 (0.141)						
log export/GDP × above top 25pc GDP 1850			0.161 (0.0691)	0.157 (0.0693)				
log export/population × above mean GDP 1850					0.185 (0.114)	0.189 (0.112)		
log export/population × above top 25pc GDP 1850							0.117 (0.0532)	0.110 (0.0512)
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>F</i>	5.385	5.404	14.33	14.71	4.026	4.012	9.954	11.17
Observations	318	318	318	318	318	318	318	318
Weighted	No	Yes	No	Yes	No	Yes	No	Yes

Notes: The table reports 2SLS estimates. The unit of observation is country-year. The dependent variable is the log of per capita GDP. Above mean GDP 1850 (above top 25 pc GDP 1850) is a dummy equal to 1 if the per capita GDP in the country was above the average (the top twenty-fifth percentile) per capita GDP across all countries in the sample in 1850. Observations are un-weighted in columns 1, 3, 5, and 7 and weighted by the log-population of the country in 1860 in columns 2, 4, 6, and 8. *F* is the Kleiberg-Paap Wald *F*-statistics for weak identification. Standard errors (reported in parentheses) are two-way clustered (country and year).

and 1939. However, to the best of my knowledge, the current study is the first to document a negative causal effect. Several authors (Lewis 1978; Williamson 2008, 2011; Acemoglu, Johnson, and Robinson 2005; and Galor and Mountford 2008) have related the first wave of trade globalization to the Great Divergence, a term coined by Samuel Huntington to describe the nineteenth century process by which the Western world overcame pre-modern growth constraints and emerged as the most powerful and wealthy world civilization of the time. Table 8 examines the relationship between trade and economic divergence.²⁰ The results are striking. An exogenous increase in international trade produced different effects depending on the initial levels of economic development: it was highly detrimental in countries characterized by a per capita GDP below the top twenty-fifth percentile in 1850, while it had a very limited negative impact on the economic performance of rich countries (see columns 3, 4, 7, and 8).

Between 1850 and 1900, the export-to-GDP ratio increased at a yearly rate of 1.3 percent, while per capita GDP increased at a rate of 1.1 percent (in the 26 countries for which per capita GDP estimates are available both in 1850 and in 1900). However, this average increase in per capita GDP masks important differences as rich countries (the top quarter in 1850 in terms of per capita GDP) grew at a yearly rate of 1.3 percent, while the rest grew at a yearly rate of 0.9 percent. Therefore,

²⁰In this table, the Kleiberg-Paap Wald *F*-statistics for weak identification is below 10 in 4 out of 8 columns. Although the critical values of the Kleiberg-Paap *F*-statistics have not been tabulated, we follow Bazzi and Clemens (2013) and compare them to the critical value of the Cragg-Donald *F*-statistics reported in Stock and Yogo (2005). In this case, the critical value for a 5 percent test of maximal size of 15 percent in the case of two endogenous variables and two excluded variables is 4.58.

the gap in the growth rate between the top quarter and the rest was approximately 0.4 percent. According to the estimates in Table 8, the first wave of trade globalization was responsible for at least 55 percent ($0.109 \times 0.013/0.004$) of this gap. The estimates in Table 8 confirm that international trade was the main force behind the Great Divergence. Moreover, they tell us that to industrialize and benefit from trade, it was not enough to start as rich because, on average, rich countries did not lose much from trade, but they also did not benefit.

The question that naturally follows is whether certain countries actually benefited from trade. In the following tables, I turn to the channels through which international trade may have affected economic development. I view this analysis as preliminary and exploratory: with a sample of only 37 countries, it is very difficult to pin down the precise channels and mechanisms through which trade affects development with a reasonable degree of certainty. My strategy here is simply to investigate whether the data are consistent with the view that the impact of trade on economic development is mediated by the quality of the local institutions and their role in shaping comparative advantages. Table 9 tests the hypothesis that trade had a differential impact on development depending on the quality of the local institutions. The basic estimating equation is as follows:

$$(9) \quad \log(Y_{it}) = \alpha_0 \log T_{it} + \alpha_1 \log T_{it} \cdot Exec_i + \gamma_i + \gamma_t + v_{it},$$

where $Exec_i$ is a measure of the constraints in the year 1860 on the decision-making power of the chief executives. This variable captures whether the executive power is constrained by checks and balances and the rule of law. The first stage of the 2SLS estimates is given by the following system of equations:

$$(10) \quad \log T_{it} = \theta_{11} \log PT_{it} + \theta_{12} \log PT_{it} \cdot Exec_i + \gamma_i + \gamma_t + \varepsilon_{1it}$$

$$(11) \quad \log T_{it} \cdot I(Exec_i) = \theta_{21} \log PT_{it} + \theta_{22} \log PT_{it} \cdot Exec_i + \gamma_i + \gamma_t + \varepsilon_{2it}.$$

The 2SLS estimates in Table 9 confirm the view that institutions are crucial to capturing the benefits of international trade. An exogenous doubling in the export-to-GDP ratio reduced per capita GDP growth rates by more than one-third in countries characterized by an executive power with unlimited authority ($Exec_i = 1$) while increasing them by almost one-tenth in countries, in which the executive power was responding to several accountability groups ($Exec_i = 7$). Conversely, the impact of trade on population does not seem to depend on institutions. (Table A.8 in the online Appendix confirms these results when using per capita exports as a measure of trade openness). The fact that only countries with inclusive institutions were able to benefit from trade does not imply a causal link. A strong argument against causality is that the estimates in Table 9 might simply be capturing the causal link between the colonization process and development. Table A.9 in the online Appendix tests whether international trade had an heterogeneous impact depending on the colonial status of the country. Unfortunately, the instruments turn out to be very weak in this particular table. With this limitation, the 2SLS estimates show that trade did not have a differential impact between colonies and independent states on per capita GDP, population, and urbanization rates. Another potential argument

TABLE 9—TRADE AND DEVELOPMENT: THE ROLE OF LOCAL INSTITUTIONS

	Per capita GDP			Population		
	(OLS) (1)	(2SLS) (2)	(2SLS) (3)	(OLS) (4)	(2SLS) (5)	(2SLS) (6)
log export/GDP	−0.178 (0.0412)	−0.398 (0.119)		−0.322 (0.112)	−1.357 (0.289)	
log export/population			−0.292 (0.114)			−1.349 (0.238)
log export/GDP × exec constraints	0.0295 (0.00973)	0.0675 (0.0208)		0.0274 (0.0161)	0.123 (0.0575)	
log export/population exec constraints			0.0412 (0.0114)			0.101 (0.0322)
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
<i>F</i>		11.98	10.18		11.98	10.18
Observations	344	344	344	344	344	344
Weighted	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports OLS and 2SLS estimates. The unit of observation is country-year. The dependent variable is either the log of per capita GDP or population. Exec constraints is the score in the POLITY IV variable constraints on the executive in 1860. The excluded instruments are constructed according to equations (10) and (11). Observations are weighted by the log-population of the country in 1860. *F* is the Kleiberg-Paap Wald *F*-statistics for weak identification. Standard errors (reported in parentheses) are two-way clustered (country and year).

against causality is that institutions might reflect differences in the initial level of either economic development or specialization in agricultural versus nonagricultural sectors. However, we have already seen that rich countries did not benefit, on average, from the first wave of trade globalization, while the estimates reported in the online Appendix (see Table A.10) show that countries that were specialized in agriculture at the beginning of the period experienced a similar impact of trade as countries that were not.

Why would we expect institutions to be crucial to benefiting from trade? A common argument is that a country with “good” institutions will suffer less from the hold-up underinvestment problem in industries that intensively rely on relationship-specific assets (for a complete review, see Nunn and Trefler 2014). In this sense, good institutions are a crucial source of comparative advantage in nonagricultural sectors, in which the hold-up problem is more binding. Table 10 shows that this was indeed the case in the second half of the nineteenth century. An exogenous increase in the exposure to international trade increased the share of exports in nonagricultural products and the share of the population living in large cities only in those countries characterized by stronger constraints on the executive power.

More specifically, in those countries characterized by the best institutions in terms of accountability of the executive power ($Exec_i = 7$), an exogenous increase in the export-to-GDP ratio by 1 percent produced an increase in the share of exports in nonagricultural products on the order of 12 percent (this increase is statistically significant at a 10 percent statistical confidence), while it had no impact in autocracies. Moreover, while the impact of a trade shock on urbanization rates was large and negative in autocracies, it was positive (although not statistically significant) in countries characterized by stronger constraints on the executive.

TABLE 10—TRADE, IDENTIFICATION, AND URBANIZATION: THE ROLE OF LOCAL INSTITUTIONS

	Share nonagricultural exports		Urban pop (>50,000)		Urban pop (>100,000)	
	(OLS) (1)	(2SLS) (2)	(OLS) (3)	(2SLS) (4)	(OLS) (5)	(2SLS) (6)
log export/GDP	−0.0313 (0.0461)	0.0156 (0.0849)	−0.0312 (0.0129)	−0.115 (0.0555)	−0.0372 (0.0150)	−0.127 (0.0680)
log export/GDP × exec constraints	0.0140 (0.00700)	0.0149 (0.0126)	0.00499 (0.00165)	0.0173 (0.00736)	0.00474 (0.00201)	0.0190 (0.00702)
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	153	153	344	344	344	344
<i>F</i>		3.286		11.98		11.98
Weighted	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports 2SLS estimates. The unit of observation is country-year. The dependent variable is either the log of the share of nonagricultural exports (columns 1 and 2), or the log of the population share living in cities with more than 50,000 citizens (columns 3 and 4), or 100,000 citizens (columns 5 and 6). Exec constraints is the score in the POLITY IV variable constraints on the executive in 1860. Observations are weighted by the log-population of the country in 1860. *F* is the Kleiberg-Paap Wald *F*-statistics for weak identification. Standard errors (reported in parentheses) are two-way clustered (country and year).

The finding that trade could have been detrimental, at least in the short run, in those countries that rely on worse institutions and, therefore, specialize in agriculture, is in line with the theoretical predictions of a large class of models in the new economic geography paradigm. In these models, positive externalities from producing manufactures imply increasing returns to scale in the secondary sector, while traditional agriculture obeys constant returns. A reduction in trade costs stimulates growth in those countries that specialize in manufacturing and can benefit from the increasing returns in this sector, while it may be harmful in the short run in countries that deindustrialize.

V. Conclusions

What factors drove globalization in the late nineteenth century? How did the rise in international trade affect economic development? This work addresses these two questions using new data and a novel identification strategy. I found that (i) the adoption of the steamship had a major impact on patterns of international trade worldwide; (ii) only a small number of countries, characterized by more inclusive institutions, benefited from trade integration; and (iii) globalization was the major driver of the economic divergence between the rich and the poor portions of the world in the years 1850–1900.

These results are important both for researchers and for policymakers.

For researchers, this paper presents the first empirical study to identify the effects of the steamship on trade and development. Moreover, researchers will be able to exploit a new source of variation in international trade, that is exogenous with respect to economic development, for studying the impact of trade on other economic/social outcomes, such as technology diffusion or conflicts.

The use of the term “globalization” has become commonplace in these last years; however, the increasing interconnection that we observe in the world today is not a

new phenomenon. The late nineteenth century is an ideal testing ground in which to observe the effects that globalization can have on economic development. In this study, I showed that the increase in international trade had heterogeneous effects on local economic development (actually, these effects were negative for the majority of countries) and increased inequality across nations. Policymakers who are willing to learn from history are advised to consider that a reduction in trade barriers across countries does not automatically produce large positive effects on economic development.

REFERENCES

- Acemoglu, Daron, and Simon Johnson.** 2005. "Unbundling Institutions." *Journal of Political Economy* 113 (5): 949–95.
- Acemoglu, Daron, Simon Johnson, and James A. Robinson.** 2001. "The Colonial Origins of Comparative Development: An Empirical Investigation." *American Economic Review* 91 (5): 1369–401.
- Acemoglu, Daron, Simon Johnson, and James Robinson.** 2005. "The Rise of Europe: Atlantic Trade, Institutional Change, and Economic Growth." *American Economic Review* 95 (3): 546–79.
- Anderson, James E., and Eric van Wincoop.** 2003. "Gravity with Gravitas: A Solution to the Border Puzzle." *American Economic Review* 93 (1): 170–92.
- Angier, E. A. V.** 1920. *Fifty Years' Freights: 1869–1919*. London: Fairplay.
- Baldwin, Richard E., Philippe Martin, and Gianmarco I. P. Ottaviano.** 2001. "Global Income Divergence, Trade, and Industrialization: The Geography of Growth Take-Offs." *Journal of Economic Growth* 6 (1): 5–37.
- Banks, Arthur S., and Kenneth A. Wilson.** 2013. *Cross-National Time-Series Data Archive*. Jerusalem: Databanks International.
- Barbieri, Katherine.** 1996. "Economic Interdependence: A Path to Peace or a Source of Interstate Conflict?" *Journal of Peace Research* 33 (1): 29–49.
- Bazzi, Samuel, and Michael A. Clemens.** 2013. "Blunt Instruments: Avoiding Common Pitfalls in Identifying the Causes of Economic Growth." *American Economic Journal: Macroeconomics* 5 (2): 152–86.
- Bolt, Jutta, and Jan Luiten van Zanden.** 2014. "The Maddison Project: Collaborative Research on Historical National Accounts." *Economic History Review* 67 (3): 627–51.
- Crafts, Nicholas, and Anthony Venables.** 2003. "Globalization in History: A Geographical Perspective." In *Globalization in Historical Perspective*, edited by Michael D. Bordo, Alan M. Taylor, and Jeffrey G. Williamson, 323–69. Chicago: University of Chicago Press.
- Eaton, Jonathan, and Samuel Kortum.** 2002. "Technology, Geography, and Trade." *Econometrica* 70 (5): 1741–79.
- Engerman, Stanley L., and Kenneth L. Sokoloff.** 1994. "Factor Endowments: Institutions, and Differential Paths of Growth Among New World Economies: A View from Economic Historians of the United States." National Bureau of Economic Research Working Paper 066.
- Estevadeordal, Antoni, Brian Frantz, and Alan M. Taylor.** 2003. "The Rise and Fall of World Trade, 1870–1939." *Quarterly Journal of Economics* 118 (2): 359–407.
- Faber, Benjamin.** 2014. "Trade Integration, Market Size, and Industrialization: Evidence from China's National Trunk Highway System." *Review of Economic Studies* 81 (3): 1046–70.
- Federico, Giovanni, and Karl Gunnar Persson.** 2007. "Market Integration and Convergence in the World Wheat Market, 1800–2000." In *The New Comparative Economic History: Essays in Honor of Jeffrey G. Williamson*, edited by Tim Hatton, Kevin O'Rourke, and Alan M. Taylor, 87–114. Cambridge, MA: MIT Press.
- Ferguson, Niall, and Moritz Schularick.** 2006. "The Empire Effect: The Determinants of Country Risk in the First Age of Globalization, 1880–1913." *Journal of Economic History* 66 (2): 283–312.
- Feyrer, James.** 2009a. "Distance, Trade, and Income: The 1967 to 1975 Closing of the Suez Canal as a Natural Experiment." National Bureau of Economic Research Working Paper 15557.
- Feyrer, James.** 2009b. "Trade and Income: Exploiting Time Series in Geography." National Bureau of Economic Research Working Paper 14910.
- Flandreau, Marc, and Mathilde Maurel.** 2001. "Monetary Union, Trade Integration, and Business Cycles in 19th Century Europe: Just Do It." Center for Economic Policy Research Discussion Paper 3087.
- Fletcher, M. E.** 1958. "The Suez Canal and World Shipping, 1869–1914." *Journal of Economic History* 18 (4): 556–73.

- Frankel, Jeffrey A., and David Romer. 1999. "Does Trade Cause Growth?" *American Economic Review* 89 (3): 379–99.
- Frieden, Jeffrey A. 2007. *Global Capitalism: Its Fall and Rise in the Twentieth Century*. New York: W. W. Norton & Company.
- Galor, Oded, and Andrew Mountford. 2008. "Trading Population for Productivity: Theory and Evidence." *Review of Economic Studies* 75 (4): 1143–79.
- Graham, Gerald S. 1956. "The Ascendancy of the Sailing Ship 1850–85." *Economic History Review* 9 (1): 74–88.
- Jacks, David S. 2006. "What Drove 19th Century Commodity Market Integration?" *Explorations in Economic History* 43 (3): 383–412.
- Jacks, David S., Christopher M. Meissner, and Dennis Novy. 2011. "Trade Booms, Trade Busts, and Trade Costs." *Journal of International Economics* 83 (2): 185–201.
- James, Harold. 2001. *The End of Globalization: Lessons from the Great Depression*. Cambridge, MA: Harvard University Press.
- Knauerhase, Ramon. 1968. "The Compound Steam Engine and Productivity Changes in the German Merchant Marine Fleet, 1871–1887." *Journal of Economic History* 28 (3): 390–403.
- Krugman, Paul. 1991. "Increasing Returns and Economic Geography." *Journal of Political Economy* 99 (3): 483–99.
- Krugman, Paul, and Anthony Venables. 1995. "Globalization and the Inequality of Nations." *Quarterly Journal of Economics* 110 (4): 857–80.
- La Porta, Rafael, Florencio Lopez-de-Silanes, Andrei Shleifer, and Robert W. Vishny. 1997. "Legal Determinants of External Finance." *Journal of Finance* 52 (3): 1131–50.
- Lampe, Markus. 2008. "Bilateral Trade Flows in Europe, 1857–1875: A New Dataset." In *Research in Economic History*, Vol. 26, edited by Alexander J. Field, William A. Sundstrom, and Gregory Clark, 81–155. Bingley, UK: Emerald Group Publishing.
- Levchenko, Andrei A. 2007. "Institutional Quality and International Trade." *Review of Economic Studies* 74 (3): 791–819.
- Lewis, William Arthur. 1978. *The Evolution of the International Economic Order*. Princeton, NJ: Princeton University Press.
- Lopez-Cordova, J. Ernesto, and Christopher M. Meissner. 2003. "Exchange-Rate Regimes and International Trade: Evidence from the Classical Gold Standard Era." *American Economic Review* 93 (1): 344–53.
- Maddison, Angus. 1995. *Monitoring the World Economy, 1820–1992*. Paris: OECD.
- Maddison, Angus. 2004. *The World Economy: Historical Statistics*. Paris: OECD Development Centre.
- Matsuyama, Kiminori. 1992. "Agricultural Productivity, Comparative Advantage, and Economic Growth." *Journal of Economic Theory* 58 (2): 317–34.
- Mitchell, Brian. 2007. *International Historical Statistics: 1750–2005: Europe*. Basingstoke, UK: Palgrave Macmillan.
- Mitchener, Kris James, and Marc Weidenmier. 2008. "Trade and Empire." *Economic Journal* 118 (533): 1805–34.
- Morton, Fiona Scott. 1997. "Entry and Predation: British Shipping Cartels 1879–1929." *Journal of Economics and Management Strategy* 6 (4): 679–724.
- North, Douglass C. 1981. *Structure and Change in Economic History*. New York: W. W. Norton & Company.
- North, Douglass C. 1990. *Institutions, Institutional Change and Economic Performance*. Cambridge, UK: Cambridge University Press.
- North, Douglass C., and Robert Paul Thomas. 1973. *The Rise of the Western World: A New Economic History*. Cambridge, UK: Cambridge University Press.
- Nunn, Nathan. 2009. "The Importance of History for Economic Development." *Annual Review of Economics* 1 (1): 65–92.
- Nunn, Nathan, and Daniel Treffer. 2014. "Domestic Institutions as a Source of Comparative Advantage." In *Handbook of International Economics*, Volume 4, edited by Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff, 263–315. Amsterdam: North Holland.
- O'Brien, Patrick. 1982. "European Economic Development: The Contribution of the Periphery." *Economic History Review* 35 (1): 1–18.
- O'Rourke, Kevin H., and Jeffrey G. Williamson. 1999. *Globalization and History: The Evolution of a Nineteenth-Century Atlantic Economy*. Cambridge, MA: MIT Press.
- Pamuk, Sevket. 2010. *The Ottoman Empire and European Capitalism, 1820–1913: Trade, Investment and Production*. Cambridge, UK: Cambridge University Press.
- Pascali, Luigi. 2017. "The Wind of Change: Maritime Technology, Trade, and Economic Development: Dataset." *American Economic Review*. <https://doi.org/10.1257/aer.20140832>.

- Podolny, Joel M., and Fiona M. Scott Morton.** 1999. "Social Status, Entry and Predation: The Case of British Shipping Cartels 1879–1929." *Journal of Industrial Economics* 47 (1): 41–67.
- Rodriguez, Francisco, and Dani Rodrik.** 2001. "Trade Policy and Economic Growth: A Skeptic's Guide to the Cross-National Evidence." In *NBER Macroeconomics Annual*, Vol. 15, edited by Ben S. Bernanke and Kenneth Rogoff, 261–338. Cambridge, MA: National Bureau of Economic Research.
- Staiger, Douglas, and James H. Stock.** 1997. "Instrumental Variables Regression with Weak Instruments." *Econometrica* 65 (3): 557–86.
- Stock, James H., and Motohiro Yogo.** 2005. "Testing for Weak Instruments in Linear IV Regression." In *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*, edited by Donald W. K. Andrews and James H. Stock, 80–108. New York: Cambridge University Press.
- Walton, Gary M.** 1970. "Productivity Change in Ocean Shipping After 1870: A Comment." *Journal of Economic History* 30 (2): 435–41.
- Williamson, Jeffrey G.** 2008. "Globalization and the Great Divergence: Terms of Trade Booms and Volatility in the Poor Periphery 1782–1913." National Bureau of Economic Research Working Paper 13841.
- Williamson, Jeffrey G.** 2011. *Trade and Poverty: When the Third World Fell Behind*. Cambridge, MA: MIT Press.